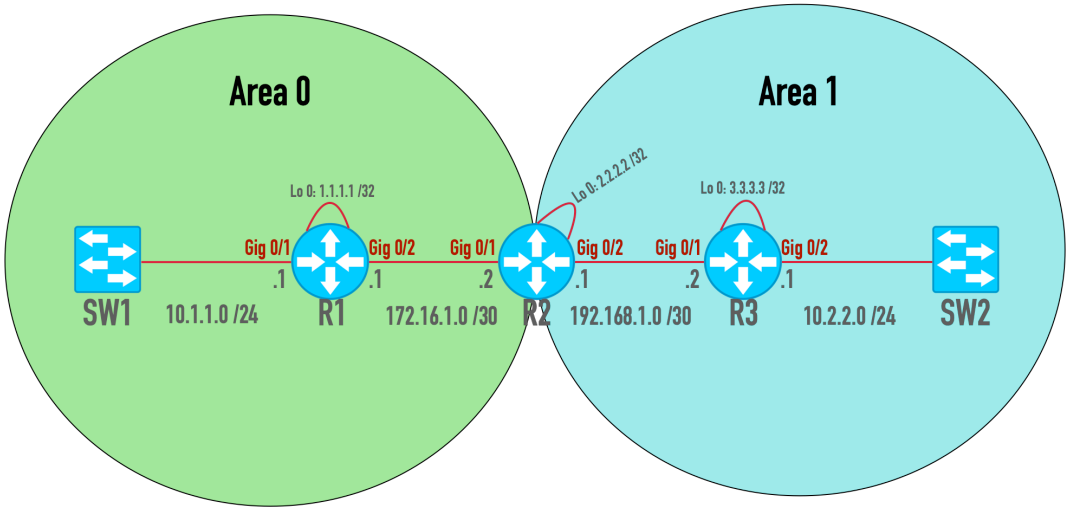
OSPF Route Filtering

Topology



Initial Configuration Commands

R1:

```
enable
conf t
host R1
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
exit
no ip-domain lookup
int gig 0/1
ip address 10.1.1.1 255.255.255.0
no shut
int gig 0/2
ip address 172.16.1.1 255.255.255.252
no shut
int lo 0
```

Copyright 2024, Kevin Wallace Training

```
ip address 1.1.1.1 255.255.255.255
end
copy run star
conf t
router ospf 1
network 10.1.1.0 0.0.0.255 area 0
network 1.1.1.1 0.0.0.0 area 0
network 172.16.1.0 0.0.0.3 area 0
exit
router ospf 1
passive-interface gig 0/1
end
```

R2:

```
enable
conf t
host R2
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
exit
no ip-domain lookup
int gig 0/1
ip address 172.16.1.2 255.255.255.252
no shut
int gig 0/2
ip address 192.168.1.1 255.255.255.252
no shut
int lo 0
ip address 2.2.2.2 255.255.255.255
end
copy run star
conf t
router ospf 2
network 172.16.1.0 0.0.0.3 area 0
network 2.2.2.2 0.0.0.0 area 1
network 192.168.1.0 0.0.0.3 area 1
end
```

Copyright 2024, Kevin Wallace Training

R3:

```
enable
conf t
host R3
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
exit
no ip-domain lookup
int gig 0/1
ip address 192.168.1.2 255.255.255.252
no shut
int gig 0/2
ip address 10.2.2.1 255.255.255.0
no shut
int lo 0
ip address 3.3.3.3 255.255.255.255
end
copy run star
conf t
router ospf 1
network 0.0.0.0 255.255.255.255 are 1
end
```

(NOTE: The base configuration of this lab should be the final configuration we have at the end of our OSPFv2 Basic Configuration lab.)

Lab Tasks

1. Let's see what routes we have knowledge of on R1.
2. On R2 (a.k.a our Area Border Router), let's create a prefix list that is going to deny 2.2.2.2 /32 and 3.3.3.3 /32, but permit everybody else. Name our prefix list "NO-LOOPBACKS".
3. On R2, let's create a filter list based on our prefix list "NO-LOOPBACKS". We want to create that filter list in area 0, going in the inbound direction.

Copyright 2024, Kevin Wallace Training

4. On R1, let's confirm that we no longer know about 2.2.2.2 /32 and 3.3.3.3 /32 in our routing table.
5. On R1, let's create a prefix list named "FILTER_10.2.2.0" to deny the 10.2.2.0 /24 network from being injected into the routing table. Then, let's permit everyone else. After that, let's apply that to our routing process by using a distribution list.
6. On R1, let's see if 10.2.2.0 /24 is in our OSPF database.
7. On R1, let's see if 10.2.2.0 /24 shows up in our ip routing table.

Solution

Step 1: Let's see what routes we have knowledge of on R1.

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
C        1.1.1.1 is directly connected, Loopback0
      2.0.0.0/32 is subnetted, 1 subnets
O IA     2.2.2.2 [110/2] via 172.16.1.2, 00:49:22, GigabitEthernet0/2
      3.0.0.0/32 is subnetted, 1 subnets
O IA     3.3.3.3 [110/3] via 172.16.1.2, 00:48:55, GigabitEthernet0/2
      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C        10.1.1.0/24 is directly connected, GigabitEthernet0/1
L        10.1.1.1/32 is directly connected, GigabitEthernet0/1
O IA     10.2.2.0/24 [110/3] via 172.16.1.2, 00:48:55, GigabitEthernet0/2
      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.16.1.0/30 is directly connected, GigabitEthernet0/2
L        172.16.1.1/32 is directly connected, GigabitEthernet0/2
      192.168.1.0/30 is subnetted, 1 subnets
O IA     192.168.1.0 [110/2] via 172.16.1.2, 00:49:22, GigabitEthernet0/2
```

(##Notice how we can see the loopback addresses of each of our routers.)

Step 2: On R2 (a.k.a our Area Border Router), let's create a prefix list that is going to deny 2.2.2.2 /32 and 3.3.3.3 /32, but permit everybody else. Name our prefix list "NO-LOOPBACKS".

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
```

Copyright 2024, Kevin Wallace Training

```
R1(config)#ip prefix-list NO-LOOPBACKS seq 10 deny 2.2.2.2/32
R1(config)#ip prefix-list NO-LOOPBACKS seq 20 deny 3.3.3.3/32
R1(config)#ip prefix-list NO-LOOPBACKS seq 30 permit 0.0.0.0/0 le 32
```

(##Le 32 means that our subnet length for our permitted addresses is less than or equal to 32, which encompasses all subnet mask lengths.)

Step 3: On R2, let's create a filter list based on our prefix list "NO-LOOPBACKS". We want to create that filter list in area 0, going in the inbound direction.

```
R1(config)#router ospf 2
R1(config-router)#area 0 filter-list prefix NO-LOOPBACKS in
R1(config-router)#end
```

Step 4: On R1, let's confirm that we no longer know about 2.2.2.2 /32 and 3.3.3.3 /32 in our routing table.

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
C        1.1.1.1 is directly connected, Loopback0
      10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C        10.1.1.0/24 is directly connected, GigabitEthernet0/1
L        10.1.1.1/32 is directly connected, GigabitEthernet0/1
O IA     10.2.2.0/24 [110/3] via 172.16.1.2, 01:49:26, GigabitEthernet0/2
      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.16.1.0/30 is directly connected, GigabitEthernet0/2
L        172.16.1.1/32 is directly connected, GigabitEthernet0/2
      192.168.1.0/30 is subnetted, 1 subnets
O IA     192.168.1.0 [110/2] via 172.16.1.2, 01:49:53, GigabitEthernet0/2
```

(##Notice how we no longer can see the loopback addresses from R2 and R3.)

Copyright 2024, Kevin Wallace Training

Step 5: On R1, let's create a prefix list named "FILTER_10.2.2.0" to deny the 10.2.2.0 /24 network from being injected into the routing table. Then, let's permit everyone else. After that, let's apply that to our routing process by using a distribution list.

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip prefix-list FILTER_10.2.2.0 seq 10 deny 10.2.2.0/24
R1(config)#ip prefix-list FILTER_10.2.2.0 seq 20 permit 0.0.0.0/0 le 32
R1(config)#router ospf 1
R1(config-router)#distribute-list prefix FILTER_10.2.2.0 in
R1(config-router)#end
```

Step 6: On R1, let's see if 10.2.2.0 /24 is in our OSPF database.

```
R1#show ip ospf database

            OSPF Router with ID (172.16.1.1) (Process ID 2)

            OSPF Router with ID (1.1.1.1) (Process ID 1)

                Router Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum
Link count
1.1.1.1         1.1.1.1         1103        0x80000006 0x00FA7A
3
2.2.2.2         2.2.2.2         1153        0x80000007 0x00F5A6
1

                Net Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum
172.16.1.2      2.2.2.2         1153        0x80000004 0x001D49

                Summary Net Link States (Area 0)

Link ID         ADV Router      Age         Seq#       Checksum
10.2.2.0        2.2.2.2         1153        0x80000004 0x00AA77
192.168.1.0     2.2.2.2         1153        0x80000004 0x008247
```

(##These are networks that we have learned from a different area, and these are based on Type 3 LSAs.)

Step 7: On R1, let's see if 10.2.2.0 /24 shows up in our ip routing table.

```
R1#show ip route
```

Copyright 2024, Kevin Wallace Training

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      1.0.0.0/32 is subnetted, 1 subnets
C        1.1.1.1 is directly connected, Loopback0
      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        10.1.1.0/24 is directly connected, GigabitEthernet0/1
L        10.1.1.1/32 is directly connected, GigabitEthernet0/1
      172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C        172.16.1.0/30 is directly connected, GigabitEthernet0/2
L        172.16.1.1/32 is directly connected, GigabitEthernet0/2
      192.168.1.0/30 is subnetted, 1 subnets
O IA     192.168.1.0 [110/2] via 172.16.1.2, 00:04:58, GigabitEthernet0/2
```

(##As you can see now, we can no longer see our 10.2.2.0 /24 network.)

Copyright 2024, Kevin Wallace Training